

MovingAverageMinMaxObserver#

[https://pytorch.org/docs/stable/generated/torch.ao.quantization.observer.Mov](https://pytorch.org/docs/stable/generated/torch.ao.quantization.observer.MovingAverageMinMaxObserver.html)

Description:

Observer module for computing the quantization parameters based on the moving average of the min and max values. This observer computes the quantization parameters based on the moving averages of minimums and maximums of the incoming tensors. The module records the average minimum and maximum of incoming tensors, and uses this statistic to compute the quantization parameters. `averaging_constant` – Averaging constant for min/max. `dtype` – dtype argument to the quantize node needed to implement the reference model spec. `qscheme` – Quantization scheme to be used

Function Signature

```
class
torch.ao.quantization.observer.MovingAverageMinMaxObserver(averaging_constant=0.01, dtype=torch.quint8,
qscheme=torch.per_tensor_affine, reduce_range=False,
quant_min=None, quant_max=None, eps=1.1920928955078125e-07,
is_dynamic=False, **kwargs)[source]#
```

Parameters

Notes & Warnings

NOTE Note Only works with torch.per_tensor_affine quantization scheme.

NOTE Note If the running minimum equals to the running maximum, the scale and zero_point are set to 1.0 and 0.

Detailed Documentation

Documentation

Observer module for computing the quantization parameters based on the moving

average of the min and max values.

This observer computes the quantization parameters based on the moving averages of minimums and maximums of the incoming tensors. The module records the average minimum and maximum of incoming tensors, and uses this statistic to compute the quantization parameters.

`averaging_constant` – Averaging constant for min/max.

`dtype` – dtype argument to the quantize node needed to implement the reference model spec.

`qscheme` – Quantization scheme to be used

`reduce_range` – Reduces the range of the quantized data type by 1 bit

`quant_min` – Minimum quantization value. If unspecified, it will follow the 8-bit setup.

`quant_max` – Maximum quantization value. If unspecified, it will follow the 8-bit setup.

`eps (Tensor)` – Epsilon value for float32, Defaults to `torch.finfo(torch.float32).eps`.

The moving average min/max is computed as follows

where $x_{min/max}$ is the running average min/max, XXX is the incoming tensor, and ccc is the `averaging_constant`.

The scale and zero point are then computed as in `MinMaxObserver`.

Only works with `torch.per_tensor_affine` quantization scheme.

If the running minimum equals to the running maximum, the scale and `zero_point` are set to 1.0 and 0.

